Debugging shapes and gradients

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Recall and choose your starting point

> Record your starting explanation for Debugging shapes and gradients before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Initialisation and reproducibility

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Training and validation curves

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training a small network

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

For this lesson, a dense layer stores one **output** unit per weight-matrix row. If an **input** has two features and the layer has three units, W has three rows and two columns, b has length 3, and one **input** x has length 2.

Compute z for each unit, then apply ReLU to each component.

Use W rows [1, 0], [0, 1], [1, -1] and b = [0, 0, 1]. For x = [1, 2], the scores are [1, 2, 0] and the activations are [1, 2, 0].

A batch stores examples in rows: **input**s [[1, 2], [3, 1]] have **shape** 2 by 2, and **output**s have **shape** 2 by 3.

Object	Declared shape	Meaning
One input x	Length 2	Two features
Weights W	3 by 2	Three units each reading two features
Bias b	Length 3	One bias per unit
Batch output	2 by 3	Two examples each with three activations

Other software may use a transposed weight convention. The key is to record and follow one **contract**. An array with the right number of values can still be wrong if features or axes are swapped.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Debugging shapes and gradients trace and record the first step you can explain without help.

Use W2: Expose a hidden bias error for Debugging **shape**s and gradients. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Debugging shapes and gradients explanation against the supplied facts before discussing it.

Use the worked case for Debugging **shape**s and gradients. Proposed explanation: “Matching **output shape** guarantees the computation”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Debug a new gradient independently

Use y = 3a + c, target 2, loss (y - 2) squared, at [a, c] = [0, 1]. A program returns gradient [-2, -6] in order [a, c].

Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Reject a shape mismatch

> Reject a shape mismatch. Apply the supplied Debugging shapes and gradients facts and keep your reasoning.

The layer expects W **shape** 3 by 2, b length 3 and x length 2. A submitted x has length 3. Should the last feature be silently ignored? State the failed **contract** and repair **process**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Expose a hidden bias error

> Expose a hidden bias error. Apply the supplied Debugging shapes and gradients facts and keep your reasoning.

Compute scores and activations for x = [3, 1] with the lesson layer, then omit the third bias. Identify the first discrepancy and the resulting activation error.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Check an update sign

> Check an update sign. Apply the supplied Debugging shapes and gradients facts and keep your reasoning.

For the diagnostic model at [1, 0], gradient [-4, -2] and rate 0.05, calculate the correct new parameters and loss. Compare with adding the gradient step.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Debug a new gradient independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Debug a new gradient independently. Apply the supplied Debugging shapes and gradients facts and keep your reasoning.

Use y = 3a + c, target 2, loss (y - 2) squared, at [a, c] = [0, 1]. A program returns gradient [-2, -6] in order [a, c].

Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

> **My reasoning and evidence**

W5 Keep independent regression checks

> Keep independent regression checks. Apply the supplied Debugging shapes and gradients facts and keep your reasoning.

A program passes only the x = [1, 2] final-activation test. Propose one intermediate-value check, one malformed-**input** check and one additional **input** that exposes the bias omission.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Debugging shapes and gradients to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled debugging **shape**s and gradients **evidence** sheet to Image-pattern Network Investigation. Show how you can check dimension **contract**s before computing a layer.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Debugging shapes and gradients

> Catch a gradient -order bug before the optimiser changes parameters. Explain the labelled step before checking the changed condition.

Catch a gradient -order bug before the optimiser changes parameters.

Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss $(y-4)^2$. Program claims gradient [-12,-8] in order [a, c].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/18

Debugging shapes and gradients

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute correct gradient

y = 2, residual -2: gradient [-8,-12], not [-12,-8].



2. Check coordinates

Step 0.1: a losses 3.24, 4.84 give -8; c losses 2.89, 5.29 give -12.



3. Apply correct update

At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

CHANGE ONE THING

The program changes its declared order to [c, a] and returns [-12,-8].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Catch a gradient -order bug before the optimiser changes parameters.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

1. Compute correct gradient

2. Check coordinates

3. Apply correct update

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply correct update At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904. Array **shape** and coordinate meaning both matter.

A vector with two entries can have the right length but assign each derivative to the wrong parameter. Label the entries a and c, then perturb one coordinate at a time to check their slopes.

Repairing the coordinate mapping is necessary before using the gradient to update the network.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The program changes its declared order to [c, a] and returns [-12,-8]. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. That vector now matches the declared coordinate order.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That vector now matches the declared coordinate order. Gradient values are meaningful only when their parameter names and array positions agree.

Repair a missing building block

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** and validation curves (ai-5/15). Goal: Compute comparable **training** and validation losses at saved checkpoints. Short practice: Can the original epoch 3 **training** loss 1 justify claiming best generalisation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

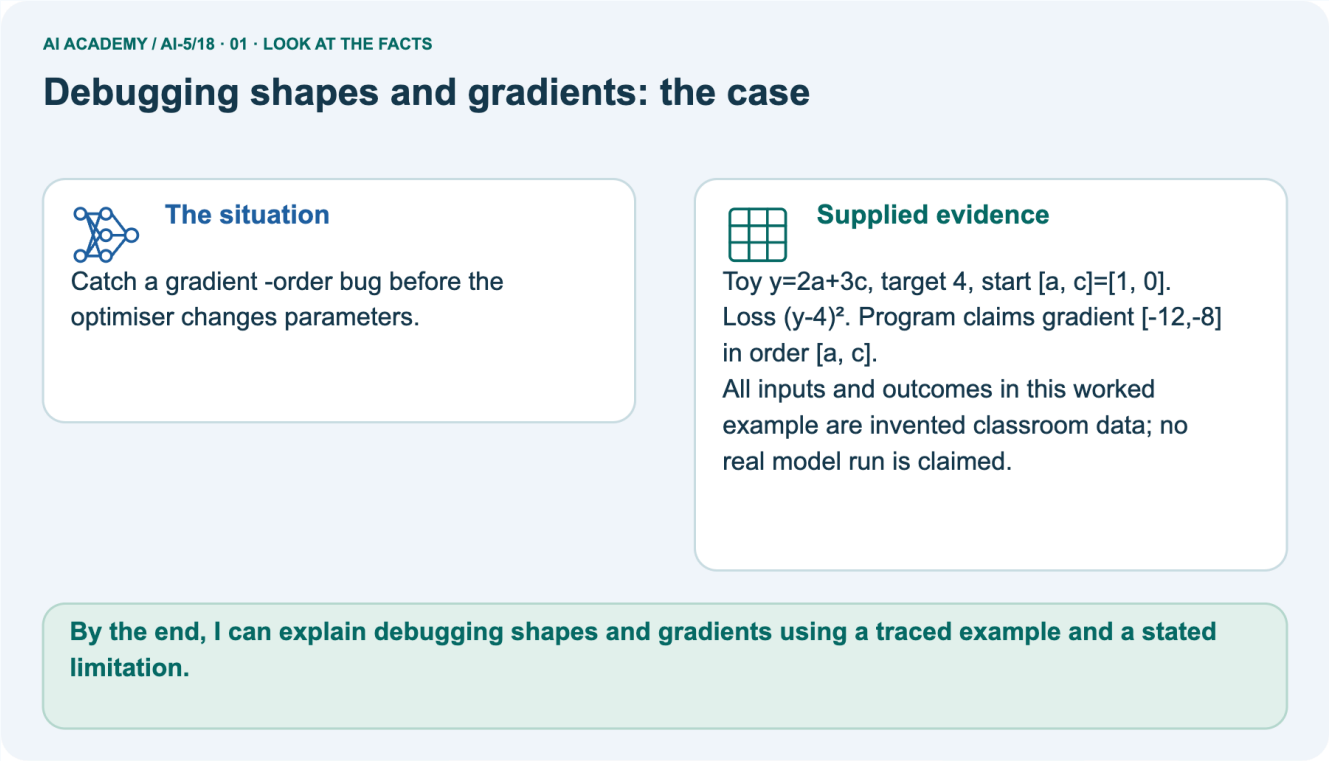
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Debugging shapes and gradients picture from its labels, then record a relevant limit.

01. Debugging shapes and gradients: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

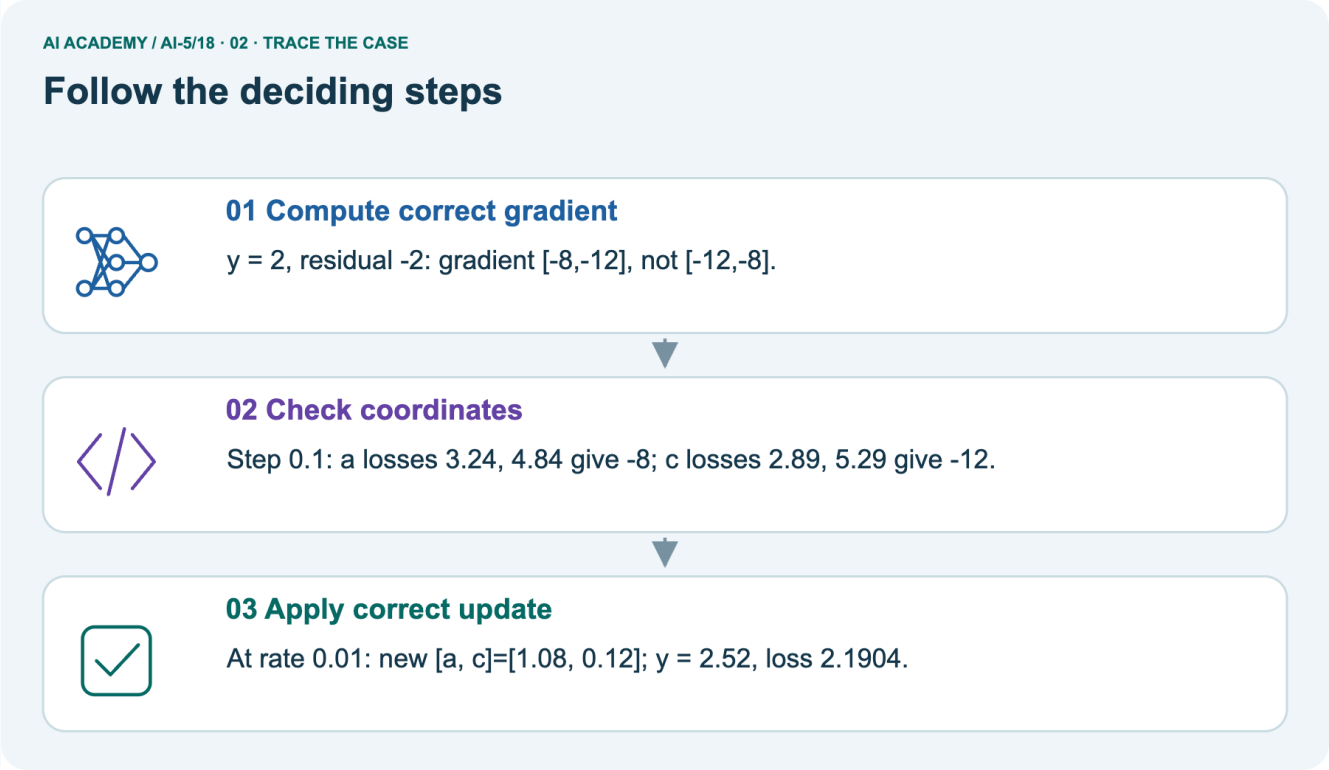
Catch a gradient -order bug before the optimiser changes parameters. Supplied facts: Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss $(y-4)^2$. Program claims gradient [-12,-8] in order [a, c].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute correct gradient: y = 2, residual -2: gradient [-8,-12], not [-12,-8]. → Check coordinates: Step 0.1: a losses 3.24, 4.84 give -8; c losses 2.89, 5.29 give -12. → Apply correct update: At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Repair from the first disagreement

AI ACADEMY / AI-5/18 · 03 · INSPECT THE LESSON MODEL

Repair from the first disagreement

Trace stage for x [1 2]	Expected result	Submitted result
Input and dimensions	x length 2 W 3 by 2 b length 3	Dimensions match
Scores before ReLU	[1, 2, 0]	[1, 2, -1]
Hidden activations	[1, 2, 0]	[1, 2, 0]

The submitted third score omits bias 1. ReLU hides the error for this input, so matching final activations are not enough. At x = [3, 1], the same omission changes the third activation from 3 to 2.

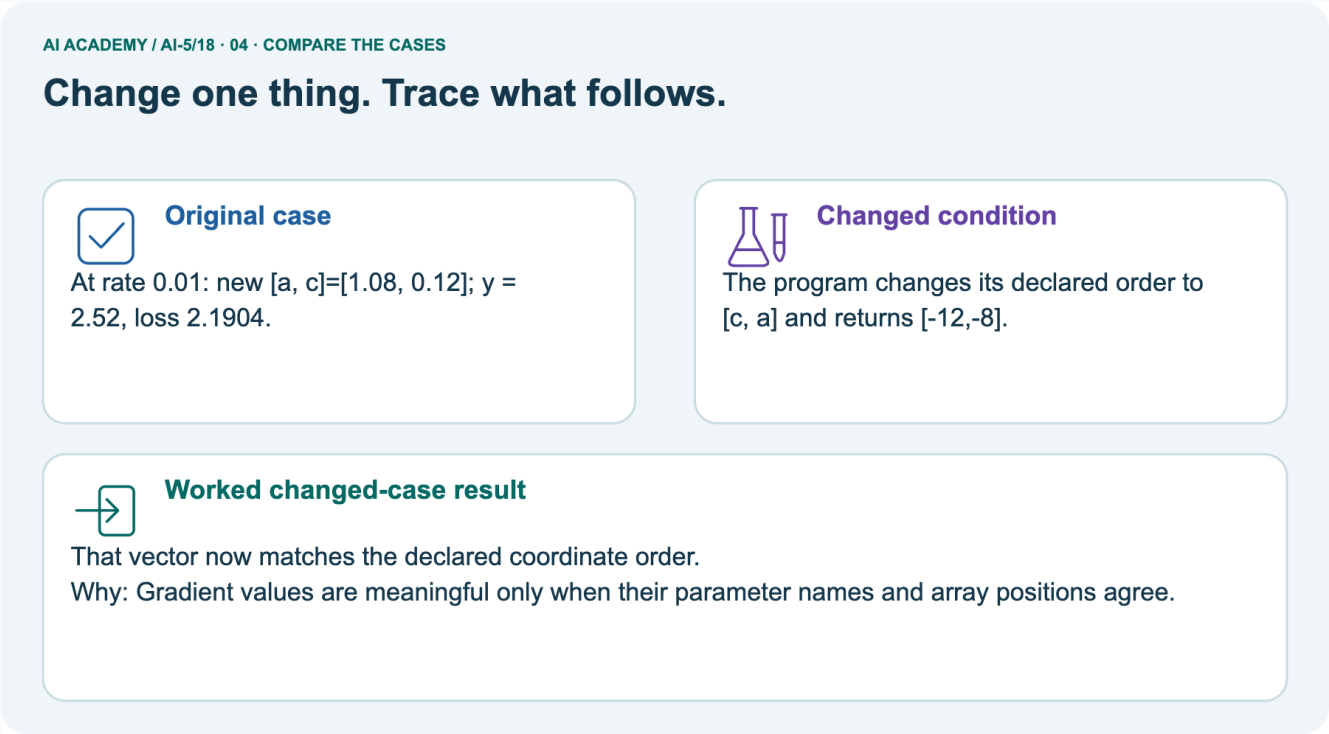
Follow the labelled picture with the matching explanation. Open the vector diagram

Repair from the first disagreement. **Input** and dimensions | x length 2 W 3 by 2 b length 3 | Dimensions match. Scores before ReLU | [1, 2, 0] | [1, 2, -1]. Hidden activations | [1, 2, 0] | [1, 2, 0]

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

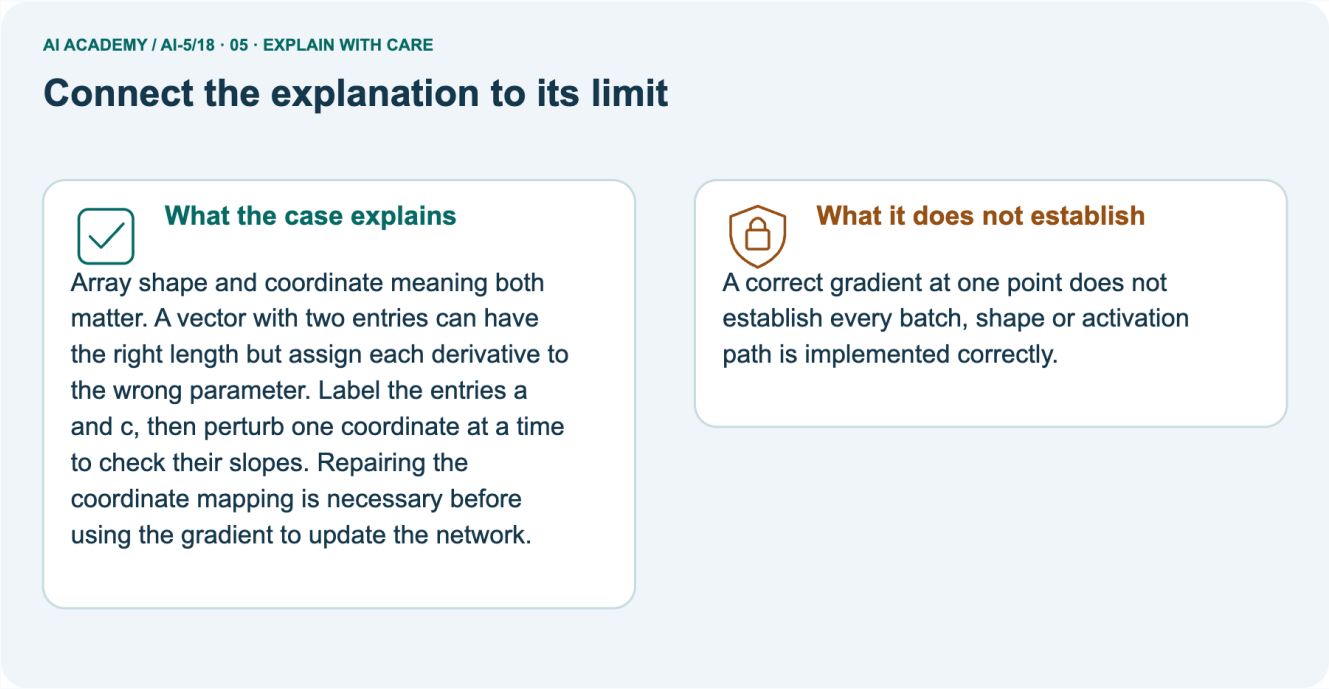
Original case: At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904. Changed condition: The program changes its declared order to [c, a] and returns [-12,-8].

Expected result: That vector now matches the declared coordinate order. Reason: Gradient values are meaningful only when their parameter names and array positions agree.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Array **shape** and coordinate meaning both matter. A vector with two entries can have the right length but assign each derivative to the wrong parameter.

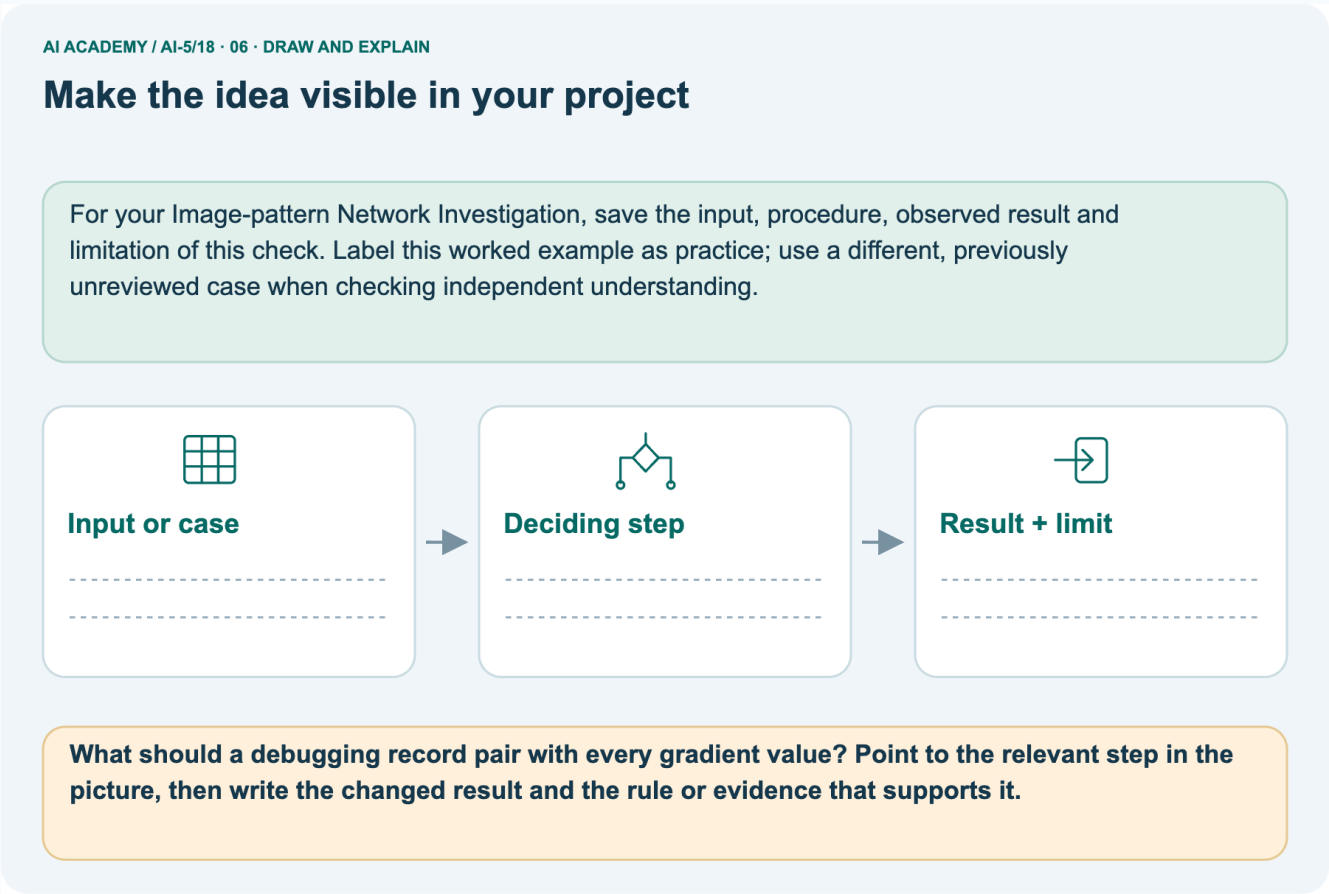
Label the entries a and c, then perturb one coordinate at a time to check their slopes. Repairing the coordinate mapping is necessary before using the gradient to update the network.

Limit: A correct gradient at one point does not establish every batch, **shape** or activation path is implemented correctly.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.